UNIVERSITY OF SCIENCE AND TECHNOLOGY OF ORAN (USTO-MB)
FACULTY OF MATHEMATICS AND COMPUTER SCIENCE — DEPARTMENT OF COMPUTER SCIENCE
MASTER 1 RSD — ADVANCED NETWORKS 1 (2026/2027)

LAB 3: HANDS-ON ANALYSIS OF THE FTP PROTOCOL

PART 1 — CONNECTION AND AUTHENTICATION

1. Launch Wireshark, start a capture on your active network interface, then apply the filter: `tcp.port == 21`

2. Open the command prompt (cmd) and type: `ftp test.rebex.net`

In Wireshark, find the first 3 TCP packets. What do the SYN, SYN-ACK and ACK packets represent?

Answer:

Then find the first FTP packet sent by the server. What is its reply code and message?

Answer:

3. In the terminal, enter the username: `anonymous` (or `ftp`)

In Wireshark, which request did the client send? Which reply (code + message) did the server return?

Request: _____ Reply: _____

4. Enter any password (for example your e-mail address).

In Wireshark, which request did the client send? Which reply (code + message) did the server return?

Request: _____ Reply: _____

5. What is the server's IP address? Which port does the server use, and which port does your machine use?

Answer:

6. In the Info column, how do you tell a client request from a server reply?

Answer:

7. Is the password readable in the `PASS` packet? Why is this a problem?

Answer:

PART 2 — NAVIGATION COMMANDS

8. Type: `pwd` — What is displayed? Which command appears in Wireshark?

Answer:

9. Type: `dir` — What is displayed? Which command appears in Wireshark?

Answer:

10. Does the file list shown by `dir` appear in the port 21 packets?

Answer:

11. Type: `cd pub/example` then `dir` — What does the server reply? Which command appears in Wireshark?

Answer:

PART 3 — DOWNLOAD AND DATA CONNECTION

12. Type: `get readme.txt` (or another file from the list). What is displayed?

Answer:

13. Still with the filter `tcp.port == 21`, find the `RETR` command. Which reply codes does the server send before and after the transfer?

Answer:

14. Does the file content appear with this filter? Why can't we see the packets of the file being transferred?

Answer:

15. Change the filter: `tcp.port == 20` (if nothing appears, use `ftp-data`). What appears now?

Answer:

16. Who sends the first SYN of this connection, the client or the server? What are the source and destination ports?

Answer:

17. Observe the data packets being sent and their ACKs. Who sends the data, who sends the ACKs? How does the sequence number change?

Answer:

18. What happens at the end of the transfer (FIN packets)? Is the port 21 connection closed too?

Answer:

19. Run another `get` or `dir`. Is the client-side port the same as the first time?

Answer:

20. Why does FTP use several ports (21, 20 and others)?

Answer:

21. Type: `bye` — What does the server reply?

Answer:

REMINDER — FTP REPLY CODES

The first digit gives the category: **1xx** preliminary reply, **2xx** success, **3xx** more information expected, **4xx** temporary error, **5xx** permanent error.

Code	Meaning
125	Data connection already open, transfer starting
150	Opening data connection
200	Command successful
220	Service ready (welcome message)
221	Closing control connection
226	Transfer complete, data connection closed
227	Entering passive mode (IP and port announced)
230	User logged in
250	File/directory action completed
257	Directory path
331	User accepted, password required
425	Cannot open data connection
426	Connection closed, transfer aborted
530	Not logged in (credentials rejected)
550	File unavailable or access denied

Credentials: server `test.rebex.net` — port 21 — username `anonymous` or `ftp` — password: any (read-only server).
To submit: Wireshark screenshots of steps 2, 4, 9, 13 and 15 + your answers.
⚠ **Rule:** Do not delete, modify or upload files on the server. Use only the demo credentials.